

and also very high frequency of STM32—the so-called F2 family. Today, with cortex M3, we have a product that is running at 120 MHz. We have extended it further in terms of performance with the latest newcomer to the STM32 family—the STM32F4. It's world's most powerful CM4 in the market today. At 168 MHz, it delivers 210 DMIPS and scores 363 CoreMark benchmark.

Can you share more details about F2 family?

The F2 line is quite interesting as besides putting advanced IP on the product, we have used one of the most advanced technologies available in the market—the e-nvm (embedded non-volatile memory). e-nvm is ST's proprietary 90nm technology process with low-power performance along with high-speed frequency (120 MHz for CM3 ST-M32F2 and 168 MHz for CM4 with STM32F4).

What applications are possible with high frequency?

There are many and there will be many more as there is always continuous demand from customers to deliver more in terms of performance. One of it is audio for sure.

Is there some certain high-quality audio equipment that this chipset solution caters to?

We are talking about a simple solution like a docking station today addressed by our CM3 F1 and F2 series. This is a family of products with different flavours. Some of these target cost-sensitive applications, some target connectivity with many communication interfaces like CAN and Ethernet, and some address industrial performance application. Still all these individual product lines remain in our proposed

STM32F1xx product portfolio. The F2 MCU (STM32F2x) is based on the best-ever-performance CM3 core. And it is now superseded by our latest device introduction – the most powerful CM4 device available in the market which you can already find in production now.

This is a family and different chipsets within a family have different applications. What is the common base that defines these to be a part of F2?

The high density for the memory, high-speed operating frequency of the CPU, fast memory access (thanks to ART accelerator proprietary IP, which allows zero-wait state access to the flash memory at 120 MHz for F2 or even 168 MHz with CM4), multi-AHB bus matrix layer for multi simultaneous master communication, and lots of advanced peripherals. Bottom line: The common base definition is maximum integration, high-density memory and performance in terms of fast speed and low power consumption. This is possible due to ST's proprietary technology which enables you to release your creativity and take maximum advantage of F2.

What are the target applications for the STM32 F4 series?

The STM32F4 can, of course, address multiple different markets or application segments as it remains a general-purpose microcontroller but with large extended performance. The single-cycle digital signal processing (DSP) instructions of the STM32F4 open the doors to the digital signal controller market that requires a high computational capability and DSP instructions for demanding applications such as high-end motor control, medical equipment and security. ●

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